

SHADOW–KLEENE SATURATION: FROM ONE TRACIAL BUG TO A FULL MF RADICAL

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ABSTRACT. We isolate a weak intermediate obstruction to operator MF: a nonidentity word that vanishes in the normalized Hilbert–Schmidt shadow of every operator-norm almost representation. Higman embedding and acylindrical small cancellation compile the normal closure of one such word onto a 2-generated finitely presented Kazhdan group. A normal-Kazhdan moving-corner detector then upgrades shadow invisibility to full operator-MF invisibility. The output has full MF radical, as does every quotient. A double-HNN compression supplies an unconditional torsion-free seed, while a threshold-free reverse-Kleene theorem reduces the computational program to constructing one shadow word.

1. THE SHADOW RESIDUAL

For dimensions d_n , put

$$\mathcal{Q}_{\text{op}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n^{\|\cdot\|_{\text{op}}} M_{d_n}(\mathbb{C}).$$

The operator-MF radical of a countable group G is

$$\text{Rad}_{\text{MF}}(G) = \bigcap_{\Theta} \ker \Theta,$$

where Θ ranges over homomorphisms from G to unitary groups of norm matrix coronas. Thus $\text{Rad}_{\text{MF}}(G) = G$ precisely when every such homomorphism is trivial.

Definition 1.1. The *operator-to-HS shadow residual* $\mathfrak{S}(G) = R_{\infty \rightarrow 2}(G)$ consists of the $w \in G$ such that, for every sequence $\phi_n : G \rightarrow \mathcal{U}(d_n)$ with

$$\|\phi_n(xy) - \phi_n(x)\phi_n(y)\|_{\text{op}} \longrightarrow 0 \quad (x, y \in G),$$

the image of w is 1 in every normalized Hilbert–Schmidt ultraproduct. A nonidentity element of $\mathfrak{S}(G)$ is a *shadow bug*.

Since normalized HS norm is bounded by operator norm, $\text{Rad}_{\text{MF}}(G) \leq \mathfrak{S}(G)$. This inclusion can be strict even for an MF group; a shadow bug is therefore not a disguised non-MF hypothesis.

Proposition 1.2 (Functoriality). *$\mathfrak{S}(G)$ is a fully invariant normal subgroup. For every homomorphism $f : G \rightarrow H$,*

$$f(\mathfrak{S}(G)) \leq \mathfrak{S}(H).$$

Consequently $w \in \mathfrak{S}(G)$ implies $\langle\langle w \rangle\rangle^G \leq \mathfrak{S}(G)$.

Proof. An operator-norm almost representation of H restricts along f to one of G . Normality follows either by applying this observation to inner automorphisms or by writing $\mathfrak{S}(G)$ as an intersection of kernels. $\square$

2. THE MOVING-CORNER DETECTOR

Theorem 2.1 (Normal-Kazhdan detector). *Let $K \trianglelefteq G$ have property (T). If $K \leq \mathfrak{S}(G)$, then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

In particular, if G has property (T) and $\mathfrak{S}(G) = G$, then $\text{Rad}_{\text{MF}}(G) = G$.

Mechanism. Suppose a norm-corona representation keeps an element of K alive. Its coordinate lifts may move only a subspace of vanishing normalized rank, so the original HS shadow need not see it. A Kazhdan Laplacian cuts out the moving projection. Normality of K makes these projections asymptotically invariant under the ambient group. Compression and polar correction give another operator-norm almost representation, now normalized by the moving rank. The Kazhdan inequality forces one element of a fixed Kazhdan set to survive in its HS ultraproduct, contradicting $K \leq \mathfrak{S}(G)$. $\square$

This detector replaces the finite spectral corner of a central involution by the moving-space projection of an infinite Kazhdan subgroup. It uses no central sign, Clifford representation, or Hilbert-hotel construction.

3. SATURATING ONE WORD

Theorem 3.1 (One-word shadow saturation). *Let G be finitely generated and recursively presented. If*

$$1 \neq w \in \mathfrak{S}(G),$$

then there is a 2-generated finitely presented acylindrically hyperbolic property-(T) group Q such that

$$\text{Rad}_{\text{MF}}(Q) = Q.$$

If the source is already finitely presented, torsion-free, and acylindrically hyperbolic, Q may be chosen torsion-free.

Proof. Effective Higman embedding [2] places G in a finitely presented group H ; functoriality preserves both the shadow property and, by injectivity, nontriviality of w . Set $E = H * F_2$ and $N = \langle\langle w \rangle\rangle^E$. The Bass–Serre action makes E non-elementary acylindrically hyperbolic with trivial finite radical. The subgroup N is infinite, normal, and hence a suitable reservoir by the acylindrical subgroup theory of Osin [4].

Take a fixed finitely presented non-elementary hyperbolic Kazhdan group H_T . Run Hull’s relative common-quotient construction with two noncommensurable loxodromics $h_1, h_2 \in N$ [3], protect w , and route every generator x_i of E by a long relation

$$x_i = W_i(h_1, h_2).$$

The resulting common quotient $E \twoheadrightarrow Q \leftarrow H_T$ is 2-generated, finitely presented, acylindrically hyperbolic, Kazhdan, and satisfies

$$q(w) \neq 1, \quad q(N) = Q.$$

Hull’s torsion clause gives the final assertion in the torsion-free case.

Because $N \leq \mathfrak{S}(E)$, functoriality yields $Q = q(N) \leq \mathfrak{S}(Q)$. Hence $\mathfrak{S}(Q) = Q$, and Theorem 2.1 gives $\text{Rad}_{\text{MF}}(Q) = Q$. $\square$

Corollary 3.2. *Every quotient of Q has full MF radical. Consequently every nontrivial quotient is non-MF and every homomorphism from Q to an MF group is trivial.*

Proof. A norm-corona representation of a quotient pulls back to one of Q . For a map $Q \rightarrow M$ with M MF, its image is both a quotient of Q and an MF subgroup, hence trivial. $\square$

If $Q = \langle a, b \mid R \rangle$ and $v \neq 1$ in Q , the finite conditions $R = 1$ and $v \neq 1$ define a nonempty clopen family of 2-marked non-MF groups: each member is a nontrivial quotient of Q . This is a finite challenge certificate for full MF failure.

4. AN UNCONDITIONAL TORSION-FREE SEED

Let P be finitely presented, torsion-free, and Kazhdan, containing

$$P_1 \times P_2 \times S \leq P,$$

where $P_i \cong P$ and S is infinite, finitely presented, nonabelian, simple, and torsion-free. Form the double HNN extension

$$E = \langle P, u_1, u_2 \mid u_i P u_i^{-1} = P_i, \ i = 1, 2 \rangle.$$

The Fournier–Facio skeleton [1] gives E finitely presented, torsion-free, and acylindrically hyperbolic. Choose $s, x \in S$ with $[s, x] \neq 1$ and put

$$t = u_1, \quad c = u_1^{-1} s u_1, \quad w = [s, x].$$

Then $[c, P] = 1$ and $tct^{-1} = s$. Britton’s lemma embeds P , so $w \neq 1$ in E . The Kazhdan compression theorem transports the asymptotic centralizer of $P_1 = tPt^{-1}$ to that of P in every normalized-HS shadow of an operator-norm almost representation. Hence s commutes there with $x \in P$, and

$$1 \neq w \in \mathfrak{S}(E).$$

Applying Theorem 3.1 directly yields a 2-generated, finitely presented, torsion-free, acylindrically hyperbolic Kazhdan group Q with $\text{Rad}_{\text{MF}}(Q) = Q$.

5. THE SHADOW-KLEENE FRONT END

Theorem 5.1 (Shadow–Kleene). *Suppose a total computable compiler sends a machine index e to a uniformly recursively presented group Γ_e and a word w_e , with*

$$M_e \downarrow \implies w_e \neq 1, \quad M_e \uparrow \implies w_e \in \mathfrak{S}(\Gamma_e).$$

Then a 2-generated finitely presented acylindrically hyperbolic Kazhdan group with full MF radical exists.

Proof. By Kleene’s recursion theorem, construct a machine which computes its own compiled pair and halts exactly when proof enumeration finds $w = 1$. It cannot halt, since the displayed HALT clause would contradict the proof it found. Nonhalting also implies $w \neq 1$: equality in a recursively presented group has a finite derivation and is recursively enumerable. The NONHALT clause now gives a nontrivial shadow bug, to which Theorem 3.1 applies. $\square$

The fixed-point machine searches proofs only. It reads no matrices, dimensions, approximation thresholds, convergence rates, or soundness moduli. The remaining computational target is therefore the qualitative implication

$$\text{NONHALT} \implies w_e \in R_{\infty \rightarrow 2}(\Gamma_e),$$

which is weaker than erasure in all HS microstates.

6. FORMALIZATION BOUNDARY

The Lean file `GroupApproximation/Sofic/OpToHSShadowResidual.lean` defines $\mathfrak{S}(G)$ and checks its normality and functoriality, the passage from one word to its normal closure, the normal-Kazhdan detector interface, and the analytic saturation endpoint `normMFResidual_eq_top_of_shadow_bug_saturation`. It also extracts the explicit HNN commutator via `FournierFacioDefectData.exists_nontrivial_opToHSShadowBug`. Effective Higman embedding, Bass–Serre/Osin suitability, and Hull’s protected common-quotient construction are the stated external group-theoretic boundary. The compression-free computational seed remains open.

REFERENCES

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